FRONTEND AI ENGINEERING TRACK · WEEK 2 · FOUNDATIONS PHASE

FE-03: The AI-Assisted Workflow Drill

Empirical comparison between unguided generation (Round 1) and specification-directed, test-verified AI pairing (Round 2) across correctness, WCAG 2.1 AA accessibility, and edge-case resilience.

CANDIDATE	ASSIGNMENT CODE	BRANCHES COMPARED	VERIFICATION RATE
KM Siam	**FE-03 (Foundations)**	**round-one vs. round-two**	**9/9 Tests (100% Pass)**

1. Executive Summary & The Core Lesson

In modern AI-accelerated software engineering, "*used AI to build it*" is not a distinguishing skill. The professional differentiator is **directing AI agents with formal specifications, architectural boundaries, edge-case constraints, and automated verification loops**.

This drill selected a core capstone-relevant component: an **AI Model Preferences & Provider Settings Form (`AISettingsForm`)**. The component was built twice under strictly separated git branches:

- **Round One (`workflow/round-one-vague`):** Triggered by a lazy, single-sentence prompt ("*build a settings form for my ai app*") with output accepted without iteration.
- **Round Two (`workflow/round-two-precise`):** Directed by a structured specification citing file dependencies, Zod schema rules, WCAG 2.1 AA accessibility attributes, and mandatory automated test assertions.

2. Concrete Branch Diff & Architectural Comparison

Executing `git diff workflow/round-one-vague..workflow/round-two-precise --stat` reveals **497 insertions and 198 deletions** across seven files, demonstrating fundamental architectural divergence:

Evaluation Dimension	Round 1: Vague Prompt (Unguided)	Round 2: Precise Prompt (Directed)
State Architecture	7 disconnected `useState` hooks; uncoordinated re-renders.	Unified `react-hook-form` with `zodResolver` and isolated renders.
Schema Validation	Ad-hoc imperative checks (`if (!apiKey)`); zero format rules.	Formal Zod schema (`aiSettingsSchema.js`) with regex, coercion, and bounds.
Accessibility (a11y)	Bare `<label>` tags without `htmlFor`; inputs lack `id`. Zero ARIA.	Strict WCAG 2.1 AA: `htmlFor/id` pairings, `aria-invalid`, `aria-describedby`, `role="alert"`.
Edge Case Defense	Accepts negative temperature (-2.5), `NaN`, blank whitespace prompts.	Explicit `.trim()`, numeric `.min(0.0).max(1.0)`, and safe HTTPS protocol check.
Verification Suite	Zero automated tests; manual manual inspection only.	Automated 9-assertion test suite (`aiSettingsForm.test.js`) passing 100%.

3. Critical AI Mistakes Caught in Round One

AI Mistake 1: Whitespace System Prompt Bypass

The unguided AI wrote `if (!systemPrompt) setError(...)`. When submitted with seventeen space characters (" "), JavaScript evaluates non-empty strings as truthy, silently accepting blank personas into the payload. Round 2 caught and fixed this via `z.string().trim().min(10)`.

AI Mistake 2: Numeric Boundaries & NaN Coercion

The unguided AI bound temperature via raw `Number(e.target.value)`. Backspacing the input resulted in `NaN`, while negative inputs like `-2.5` passed validation freely. Round 2 enforced `z.coerce.number().min(0.0).max(1.0)` and verified boundary rejection.

4. Review Effort & End-to-End Velocity Analysis

A critical takeaway from the drill is the divergence between *initial generation speed* and *end-to-end delivery velocity*:

Round 1 Velocity Breakdown

- **Prompting Time:** ~20 seconds (one sentence).
- **Generation Time:** ~10 seconds.
- **Review & Remediation Time:** ~45 minutes (fixing missing labels, ARIA attributes, NaN crashes, and adding missing tests).
- **Total End-to-End Time: ~45 minutes**

Round 2 Velocity Breakdown

- **Prompting & Spec Time:** ~3 minutes (formalizing constraints).
- **Generation Time:** ~25 seconds.
- **Review & Remediation Time:** ~2 minutes (verifying test suite output).
- **Total End-to-End Time: ~5.5 minutes (8x Faster Overall!)**

5. Three Project Rules Learned & Added to Governance

The findings from this drill were codified directly into `CLAUDE.md`, `.cursorrules`, and `.cursor/rules/frontend-ai.mdc`:

```
1. Schema-Driven Form Architecture:
   All user-facing forms must use `react-hook-form` paired with `@hookform/resolvers/zod`
   and an externalized Zod schema in `src/schemas/`. Disjointed `useState` form state
   soup and manual inline string validation (e.g. `if (!input) alert()`) are forbidden.

2. Mandatory WCAG 2.1 AA Form Accessibility:
   Every form input must be programmatically paired with a dedicated ``
   matching the control's `id`. When validation errors exist, the input must declare
   `aria-invalid={true}` and `aria-describedby` pointing to an error container carrying `role="alert"`.

3. Boundary Sanitation & Test Verification Gate:
   Text inputs must enforce `.trim()` in their Zod schema to prevent whitespace-only bypasses,
   and numeric inputs must use `z.coerce.number()` with explicit `.min()` and `.max()` boundaries.
   Every new form component must be accompanied by an automated test suite before merging.
```

6. Automated Verification Test Suite (9/9 Passed)

Running `npm test` (or `node src/round2_runner.js`) on `main` and `workflow/round-two-precise` confirms 100% test pass rate:

```
============================================================
ROUND 2: PRECISE PROMPT AUTOMATED VERIFICATION TEST SUITE
Target: AISettingsForm + aiSettingsSchema
============================================================
  ✓ PASS [1]: Valid enterprise configuration passes schema validation
  ✓ PASS [2]: Invalid API key prefix rejected by regex constraint
  ✓ PASS [3]: Short API key (< 16 characters) rejected
  ✓ PASS [4]: Negative temperature (-0.25) rejected
  ✓ PASS [5]: Temperature exceeding 1.0 (1.45) rejected
  ✓ PASS [6]: Whitespace-only system prompt rejected after .trim()
  ✓ PASS [7]: Non-HTTPS webhook URL (insecure http://) rejected
  ✓ PASS [8]: Empty webhook URL gracefully accepted as optional
  ✓ PASS [9]: AISettingsForm renders complete WCAG 2.1 AA semantic markup

============================================================
TEST SUITE RESULTS: 9/9 TESTS PASSED (100% SUCCESS RATE)
============================================================
```

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